ANGEL METAL INDIA
STRENGTH • QUALITY • TRUST

Solid and Stranded Conductor

AWG Size	Total Strands/ Strand Size	Type	Construction	Nominal Diameter		Circular Area		Approimate Weight		Nom. Break Strength		Maximum DC Resistance	
				Inches	mm	Mils	mm2	Lbs/ 1000'	Kg/ Km	Lbs	Kg	Ohms/ 1,000'	Ohms/ Km
56	1	S	Solid	0.00049	0.0124	0.24	0.00012	0.00070	0.001	0.0066	0.00	46949	154039
54	1	S	Solid	0.00062	0.0157	0.38	0.00020	0.0012	0.002	0.011	0.00	28808	94520
52	1	S	Solid	0.00078	0.0198	0.61	0.00031	0.0018	0.003	0.017	0.01	18437	60493
50	1	S	Solid	0.00099	0.0251	0.98	0.00050	0.0030	0.005	0.027	0.01	11491	37703
48	1	S	Solid	0.00124	0.0315	1.54	0.00078	0.0047	0.007	0.042	0.02	7324	24029
46	1	S	Solid	0.00157	0.0399	2.46	0.0012	0.0075	0.011	0.068	0.03	4548	14924
44	1	S	Solid	0.00200	0.0508	4.00	0.0020	0.0121	0.018	0.11	0.05	2873	9426
42	1	S	Solid	0.00250	0.0635	6.25	0.0032	0.0189	0.028	0.17	0.08	1801	5908
42	7/50	C	7/50	0.00300	0.0762	6.86	0.0035	0.0216	0.032	0.19	0.09	1891	5548
40	1	S	Solid	0.00310	0.0787	9.61	0.0049	0.0291	0.043	0.26	0.12	1152	3781
40	7/48	C	7/48	0.00370	0.0940	10.8	0.0055	0.0339	0.050	0.30	0.13	1078	3536
40	10/50	B	10/50	0.00360	0.0914	9.80	0.0050	0.0306	0.046	0.27	0.12	1172	3846
39	1	S	Solid	0.00350	0.0889	12.3	0.0062	0.0371	0.055	0.34	0.15	897	2944
38	1	S	Solid	0.00400	0.102	16.0	0.0081	0.0484	0.072	0.44	0.20	682	2237
38	7/46	C	7/46	0.00470	0.119	17.3	0.0087	0.0541	0.081	0.47	0.22	669	2196
38	10/48	B	10/48	0.00450	0.114	15.4	0.0078	0.0479	0.071	0.42	0.19	747	2451
38	16/50	B	16/50	0.00460	0.117	15.7	0.0080	0.0490	0.073	0.43	0.20	733	2404
36	1	S	Solid	0.0050	0.127	25.0	0.013	0.0757	0.11	0.69	0.31	432	1417
36	7/44	C	7/44	0.0060	0.152	28.0	0.014	0.0872	0.13	0.77	0.35	423	1387
36	10/46	B	10/46	0.0057	0.145	24.6	0.012	0.0765	0.11	0.68	0.31	464	1522
36	16/48	B	16/48	0.0057	0.145	24.6	0.012	0.0767	0.11	0.68	0.31	467	1532
36	19/48	C	19/48	0.0062	0.157	29.2	0.015	0.0920	0.14	0.80	0.36	397	1303
36	25/50	B	25/50	0.0057	0.145	24.5	0.012	0.0765	0.11	0.67	0.31	469	1538
34	1	S	Solid	0.0063	0.160	39.7	0.020	0.120	0.18	1.1	0.50	270	885
34	7/42	C	7/42	0.0075	0.191	43.8	0.022	0.136	0.20	1.2	0.55	265	869
34	10/44	B	10/44	0.0073	0.185	40.0	0.020	0.123	0.18	1.1	0.50	293	961
34	16/46	B	16/46	0.0073	0.185	39.4	0.020	0.122	0.18	1.1	0.49	290	951
34	19/46	C	19/46	0.0079	0.201	46.8	0.024	0.147	0.22	1.3	0.58	247	809
34	25/48	B	25/48	0.0072	0.183	38.4	0.019	0.120	0.18	1.1	0.48	299	980
34	40/50	B	40/50	0.0072	0.183	39.2	0.020	0.122	0.18	1.1	0.49	293	961
32	1	S	Solid	0.0080	0.203	64.0	0.032	0.194	0.29	1.8	0.80	166	545
32	7/40	C	7/40	0.0093	0.236	67.3	0.034	0.210	0.31	1.9	0.84	170	556
32	10/42	B	10/42	0.0091	0.231	62.5	0.032	0.193	0.29	1.7	0.78	184	603
32	16/44	B	16/44	0.0092	0.234	64.0	0.032	0.197	0.29	1.8	0.80	183	601
32	19/44	C	19/44	0.0100	0.254	76.0	0.039	0.237	0.35	2.1	0.95	156	511
32	25/46	B	25/46	0.0091	0.231	61.6	0.031	0.190	0.29	1.7	0.77	186	609
32	64/50	B	64/50	0.0091	0.231	62.1	0.032	0.196	0.29	1.7	0.78	183	601
30	1	S	Solid	0.0100	0.254	100	0.051	0.303	0.45	2.8	1.25	106	347
30	7/38	C	7/38	0.0120	0.305	112	0.057	0.349	0.52	3.1	1.40	100	329
30	10/40	B	10/40	0.0113	0.287	96.10	0.049	0.297	0.44	2.6	1.20	118	386
30	16/42	B	16/42	0.0116	0.295	100	0.051	0.308	0.46	2.8	1.25	115	377
30	19/42	C	19/42	0.0125	0.318	119	0.060	0.370	0.55	3.3	1.48	97.6	320
30	25/44	B	25/44	0.0116	0.295	100	0.051	0.309	0.46	2.8	1.25	117	385
30	40/46	B	40/46	0.0115	0.292	98.60	0.050	0.306	0.46	2.7	1.23	116	381
29	1	S	Solid	0.0113	0.287	128	0.065	0.387	0.58	3.5	1.59	82.7	271
29	51/46	B	51/46	0.0129	0.328	126	0.064	0.390	0.58	3.5	1.57	91.0	299

B – Bunch stranded wire. Wires are twisted without a geometric relationship to each other.

C – Concentric stranded wire. Each layer of the stranding has all strands in the same direction and position.

RB – Rope construction with Bunch stranded groups. Similar to concentric for the groups of strands. Each Bunch Stranded strands.

RC – Rope construction with Concentric stranded groups. Similar to concentric stranding for both the final stranding and each group.

S – Solid wires.

The maximum resistance values are for the wire as a single conductor. Additional allowances have to be made when the wires are cabled into a multi conductor cable.

ANGEL METAL INDIA
STRENGTH • QUALITY • TRUST

Solid and Stranded Conductor

AWG Size	Total Strands/ Strand Size	Type	Construction	Nominal Diameter		Circular Area		Approimate Weight		Nom. Break Strength		Maximum DC Resistance	
				Inches	mm	Mils	mm2	Lbs/ 1000'	Kg/ Km	Lbs	Kg	Ohms/ 1,000'	Ohms/ Km
28	1	S	Solid	0.0126	0.320	159	0.080	0.481	0.72	4.4	1.98	66.4	218
28	7/36	C	7/36	0.0150	0.381	175	0.089	0.546	0.81	4.8	2.18	63.6	209
28	19/40	C	19/40	0.0155	0.394	183	0.093	0.569	0.85	5.0	2.28	62.5	205
28	26/42	B	26/42	0.0147	0.373	163	0.082	0.501	0.75	4.5	2.03	70.6	232
28	36/44	RB	2 x 18/44	0.0141	0.358	144	0.073	0.467	0.69	4.0	1.80	85.5	280
28	40/44	B	40/44	0.0146	0.371	160	0.081	0.494	0.74	4.4	2.00	73.3	240
28	65/46	B	65/46	0.0146	0.371	160	0.081	0.497	0.74	4.4	2.00	71.4	234
28	66/46	RB	3 x 22/46	0.0188	0.478	163	0.082	0.530	0.79	4.5	2.03	73.8	242
26	1	S	Solid	0.0159	0.404	253	0.13	0.765	1.1	7.0	3.15	42.1	138
26	7/34	C	7/34	0.0189	0.480	278	0.14	0.866	1.3	7.6	3.46	39.7	130
26	10/36	B	10/36	0.0183	0.465	250	0.13	0.772	1.2	6.9	3.12	44.1	145
26	19/38	C	19/38	0.0200	0.508	304	0.15	0.947	1.4	8.4	3.79	37.0	121
26	26/40	B	26/40	0.0183	0.465	250	0.13	0.772	1.2	6.9	3.12	45.2	148
26	28/40	RB	7 x 4/40	0.0208	0.528	269	0.14	0.873	1.3	7.4	3.36	44.1	145
26	41/42	B	41/42	0.0185	0.470	256	0.13	0.790	1.2	7.0	3.20	44.8	147
26	64/44	RB	4 x 16/44	0.0203	0.516	256	0.13	0.829	1.2	7.0	3.19	48.1	158
26	65/44	B	65/44	0.0186	0.472	260	0.13	0.802	1.2	7.2	3.24	45.1	148
26	66/44	RB	3 x 22/44	0.0206	0.523	264	0.13	0.855	1.3	7.3	3.29	46.6	153
26	105/46	RB	3 x 35/46	0.0204	0.518	259	0.13	0.843	1.3	7.1	3.23	46.4	152
24	1	S	Solid	0.0201	0.511	404	0.20	1.22	1.8	11.0	5.04	26.2	85.9
24	7/32	C	7/32	0.0240	0.610	448	0.23	1.40	2.1	12.3	5.59	24.5	80.2
24	10/34	B	10/34	0.0230	0.584	397	0.20	1.23	1.8	10.9	4.95	27.5	90.3
24	16/36	B	16/36	0.0231	0.587	400	0.20	1.24	1.8	11.0	4.99	27.5	90.4
24	19/36	C	19/36	0.0250	0.635	475	0.24	1.48	2.2	13.1	5.92	23.4	76.8
24	41/40	B	41/40	0.0229	0.582	394	0.20	1.22	1.8	10.8	4.91	28.7	94.1
24	42/40	B	42 /40	0.0232	0.589	404	0.20	1.25	1.9	11.1	5.03	28.0	91.8
24	65/42	B	65/42	0.0233	0.592	406	0.21	1.25	1.9	11.2	5.07	28.3	92.7
24	100/44	RB	2 x 50/44	0.0254	0.645	400	0.20	1.30	1.9	11.0	4.99	30.8	101
24	105/44	RB	3 x 35/44	0.0260	0.660	420	0.21	1.36	2.0	11.5	5.24	29.3	96.1
24	105/44	RB	7 x 15/44	0.0260	0.660	420	0.21	1.36	2.0	11.5	5.24	29.3	96.1
22	1	S	Solid	0.0253	0.643	640	0.32	1.94	2.9	17.6	7.98	16.6	54.4
22	7/30	C	7/30	0.0300	0.762	700	0.35	2.18	3.3	19.2	8.73	15.6	51.1
22	16/34	B	16/34	0.0291	0.739	635	0.32	1.96	2.9	17.5	7.92	17.2	56.4
22	19/34	C	9/34	0.0315	0.800	754	0.38	2.35	3.5	20.7	9.40	14.6	48.0
22	26/36	B	26/36	0.0294	0.747	650	0.33	2.01	3.0	17.9	8.10	17.0	55.6
22	66/40	B	66/40	0.0291	0.739	634	0.32	1.96	2.9	17.4	7.91	17.8	58.4
22	66/40	RB	2 x 33/40	0.0320	0.813	634	0.32	2.06	1.1	17.4	7.91	18.7	61.4
22	66/40	RB	3 x 22/40	0.0320	0.813	634	0.33	2.06	3.1	17.4	7.91	18.7	61.4
22	150/44	RB	3 x 50/44	0.0311	0.790	600	0.30	1.94	2.9	16.5	7.48	20.5	67.3
22	154/44	RB	7 x 22/44	0.0315	0.800	616	0.31	2.00	3.0	16.9	7.68	20.0	65.6
22	168/44	RB	7 x 24/44	0.0329	0.836	672	0.34	2.18	3.2	18.5	8.38	18.3	60.1
20	1	S	Solid	0.0320	0.813	1024	0.52	3.10	4.6	28.1	12.80	10.3	33.9
20	7/28	C	7/28	0.0378	0.960	1111	0.56	3.32	5.2	30.5	13.90	9.77	32.0
20	10/30	B	10/30	0.0365	0.927	1000	0.51	3.09	4.6	27.5	12.50	10.8	35.4
20	19/32	C	19/32	0.0400	1.016	1216	0.62	3.79	5.6	33.4	15.20	9.01	29.6

B – Bunch stranded wire. Wires are twisted without a geometric relationship to each other.

C – Concentric stranded wire. Each layer of the stranding has all strands in the same direction and position.

RB – Rope construction with Bunch stranded groups. Similar to concentric for the groups of strands. Each Bunch Stranded strands.

RC – Rope construction with Concentric stranded groups. Similar to concentric stranding for both the final stranding and each group.

S – Solid wires.

The maximum resistance values are for the wire as a single conductor. Additional allowances have to be made when the wires are cabled into a multi conductor cable.

ANGEL METAL INDIA
STRENGTH • QUALITY • TRUST

Solid and Stranded Conductor

AWG Size	Total Strands/ Strand Size	Type	Construction	Nominal Diameter		Circular Area		Approimate Weight		Nom. Break Strength		Maximum DC Resistance	
				Inches	mm	Mils	mm2	Lbs/ 1000'	Kg/ Km	Lbs	Kg	Ohms/ 1,000'	Ohms/ Km
20	20/33	B	20/33	0.0367	0.932	1008	0.51	3.11	4.6	27.7	32.00	10.8	35.4
20	26/34	B	26/34	0.0371	0.942	1032	0.52	3.19	4.7	28.4	12.90	10.6	34.7
20	41/36	B	41/36	0.0370	0.940	1025	0.52	3.17	4.7	28.2	12.80	10.8	35.3
20	42/36	RB	7 x 6/36	0.0412	1.046	1050	0.53	3.41	5.1	28.9	33.10	11.0	36.1
20	65/38	B	65/38	0.0372	0.945	1040	0.53	3.21	1.8	28.6	33.00	10.7	35.1
20	104/40	B	104/40	0.0365	0.927	999	0.51	3.09	4.6	27.5	32.50	11.3	37.1
20	105/40	RB	7 x 15/40	0.0404	1.026	1909	0.51	3.27	4.9	27.7	12.60	11.8	38.5
20	154/42	RB	7 x 22/42	0.0394	1.001	963	0.49	3.12	4.0	26.5	2.00	12.5	41.1
20	266/44	RB	7 x 38/44	0.0414	1.05	1004	0.54	3.45	5.1	29.2	13.30	11.6	38.0
18	1	S	Solid	0.0403	1.02	1624	0.82	4.92	7.3	44.6	20.30	6.51	21.4
18	7/26	C	7/26	0.0477	1.21	1770	0.90	5.52	8.2	48.6	22.10	6.19	20.3
18	10/28	B	10/28	0.0460	1.17	1588	0.80	4.90	7.3	43.6	19.80	6.77	22.2
18	16/30	B	16/30	0.0462	1.17	1600	0.81	4.94	7.4	44.0	20.00	6.75	22.1
18	19/30	C	19/30	0.0500	1.27	1900	0.96	5.92	8.8	52.2	23.70	5.74	18.8
18	32/33	B	32/33	0.0464	1.18	1613	0.82	4.98	7.4	44.3	20.10	6.75	22.1
18	41/34	B	41/34	0.0466	1.18	1627	0.82	5.02	7.5	44.7	20.30	6.71	22.0
18	63/36	RB	7 x 9/36	0.0504	1.28	1575	0.80	5.11	7.6	43.3	19.60	7.34	24.1
18	65/36	B	65/36	0.0466	1.18	1625	0.82	5.02	7.5	44.7	20.30	6.78	22.2
18	105/38	RB	7x 15/38	0.0521	1.32	1680	0.85	5.44	8.1	46.2	20.90	6.96	22.8
18	168/40	RB	7 x 24/40	0.0510	1.30	1615	0.84	5.24	7.8	44.4	20.10	7.35	24.1
18	259/42	RB	7 x 37/42	0.0511	1.30	1619	0.82	5.24	7.8	44.5	20.20	7.45	24.4
18	413/44	RB	7 x 59/44	0.0516	1.31	1652	0.84	5.35	8.0	45.4	20.60	7.45	24.4
16	1	S	Solid	0.0508	1.29	2581	1.31	7.81	11.6	70.9	32.20	4.10	13.5
16	7/24	C	7/24	0.0603	1.53	2828	1.43	8.82	13.1	71.7	35.30	3.85	12.6
16	16/28	B	16/28	0.0582	1.48	2540	1.29	7.84	11.7	69.8	31.70	4.23	33.9
16	19/29	C	19/29	0.0565	1.44	2426	1.23	7.56	11.3	66.7	30.30	4.48	14.7
16	26/30	B	26/30	0.0589	1.50	2600	1.32	8.03	11.9	71.5	32.40	4.15	33.6
16	40/32	B	40/32	0.0584	1.48	2560	1.30	7.90	11.8	70.4	31.90	4.24	13.9
16	42/32	RB	7 x 6/32	0.0659	1.67	2688	1.36	8.71	13.0	73.9	33.50	4.24	13.9
16	50/33	B	50/33	0.0580	1.47	2521	1.28	7.78	11.6	69.3	31.40	4.32	14.2
16	65/34	B	65/34	0.0587	1.49	2580	1.31	7.96	11.8	70.9	32.20	4.23	13.9
16	105/36	B	105/36	0.0592	1.50	2625	1.33	8.11	12.1	72.2	32.70	4.20	13.8
16	105/36	RB	7 x 15/36	0.0651	1.65	2625	1.33	8.51	12.7	72.2	32.70	4.41	14.5
16	168/38	RB	7 x 24/38	0.0659	1.67	2688	1.36	8.71	13.0	73.9	33.50	4.35	14.3
16	259/40	RB	7 x 37/40	0.0634	1.61	2489	1.26	8.07	12.0	68.4	31.00	4.77	15.6
16	264/40	RB	4 x 66/40	0.0640	1.63	2537	1.29	8.23	12.2	69.7	31.60	4.68	15.3
16	280/40	RB	7 x 40/40	0.0659	1.67	2691	1.36	8.73	13.0	74.0	33.60	4.41	14.5
16	665/44	RB	7 x 95/44	0.0655	1.66	2660	1.35	8.62	12.8	73.1	33.20	4.63	15.2
16	714/44	RB	7 x 3 x 34/44	0.0741	1.88	2856	1.45	9.72	14.5	78.5	35.60	4.53	14.9
14	1	S	Solid	0.0641	1.63	4109	2.08	10.9	16.2	91.4	41.50	3.51	11.5
14	7/22	C	7/22	0.0759	1.93	4461	2.27	14.0	20.8	123	55.90	2.44	8.01
14	7/22	C	7/22	0.0759	1.93	4481	2.27	11.9	17.7	101	45.80	3.15	10.3
14	19/27	C	19/27	0.0710	1.80	3831	1.94	10.9	16.2	92.5	41.90	3.53	11.6
14	19/27	C	19/27	0.0710	1.80	3831	1.94	11.9	17.8	105	47.80	2.83	9.28
14	26/28	B	26/28	0.0742	1.88	4128	2.09	12.7	19.0	113	51.50	2.60	8.54
14	26/28	B	26/28	0.0742	1.88	4128	2.09	11.5	17.1	93.0	42.20	3.68	12.1

B – Bunch stranded wire. Wires are twisted without a geometric relationship to each other.

C – Concentric stranded wire. Each layer of the stranding has all strands in the same direction and position.

RB – Rope construction with Bunch stranded groups. Similar to concentric for the groups of strands. Each Bunch Stranded strands.

RC – Rope construction with Concentric stranded groups. Similar to concentric stranding for both the final stranding and each group.

S – Solid wires.

The maximum resistance values are for the wire as a single conductor. Additional allowances have to be made when the wires are cabled into a multi conductor cable.